MILITARY SPECIFICATION

CONVERTER, INTERFACE UNIT, CV-3507/USQ-74

This amendment forms a part of Military Specification MIL-C-28869, dated 13 January 1982, and is approved for use by the Space and Naval Warfare Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

1.1, line 2: Delete "CV-3507/USQ-74" and substitute "CV-3507A/USQ-74A".

PAGE 2

2.1.1 "STANDARDS", "MILITARY": Delete

"DoD-STD-1399, Section 300 Interface Standard For Shipboard Systems, Electric Power, Alternating Current, Section 300"

and substitute: "DoD-STD-1399, Section 300 Interface Standards For Shipboard Systems, Electric Power, Alternating Current (Metric)".

PAGE 3

2.1.2 "PUBLICATIONS" "MILITARY": Delete "NAVAL ELECTRONIC SYSTEMS COMMAND (NAVELEX)" and substitute "SPACE AND NAVAL WARFARE SYSTEMS COMMAND (SPAWAR)".

2.1.2 "PUBLICATIONS" "MILITARY": Delete "NAVAL PERSONNEL (NAVPERS)" and substitute "NAVAL MILITARY PERSONNEL COMMAND (NMPC)".

After 2.1.2, add new paragraph:

"2.2 Other publication. The following document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

ELECTRICAL INDUSTRIES ASSOCIATION (EIA)

EIA RS-310-C-77 Racks, Panels, And Associated Equipment

(Application for copies should be addressed to the Electronic Industries Association, Engineering Department, 2001 Eye Street, NW, Washington, DC 20006.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)".

The attached insertable replacement pages listed below are replacements for stipulated pages. When the new pages have been entered in the document, insert the amendment as the cover sheet to the specification.

Replacement page	Page replaced
11	11
12	12
13	13
14	14

AMSC N/A FSC 7025

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3.1, line 3: Delete "(NAVELEX publication" and substitute "(SPAWAR".

PAGE 4

FIGURE 1: Delete "J-3546/USQ-74 JACK BOX" and substitute "J-3546A/USQ-74A JACK BOX".

FIGURE 1, title Delete "CV-3507/USQ-74" and substitute "CV-3507A/USQ-74A".

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FIGURE 2, title Delete "CV-3507/USQ-74" and substitute "CV-3507/A/USQ-74A".

FIGURE 2: Delete "J-3546/USQ-74 JACK BOX" and substitute "J-3546A/USQ-74A JACK BOX".

FIGURE 2: Delete "CV-3507/USQ-74 Converter interface unit" and substitute "CV-3507A/USQ-74A".

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3.2.1.1.4.1, last line Delete "NAVELEX 0967-LP-598-1010 series" and substitute "SPAWAR 0967-LP-598-1010 series".

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3.2.1.8, line 2 Delete "(J-3546/USQ-74)" and substitute "(J-3546A/USQ-74A)".

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3.3.3.1, line 6 Delete "CV-3507/USQ-74" and substitute "CV-3507A/USQ-74A".

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3.4.1, line 2 Delete "Hertz" and substitute "hertz".

3.4.2, line 2 Delete "Section 103 of MIL-STD-1399" and substitute "DoD-STD-1399, Section 300".

3.4.4.1, lines 2 through 7: Delete and substitute:

"$I_{pin}(t) = \pm I_p(f_o)\ e^{-\pi f_o t/Q} \sin(2\pi f_o t)$ and

$V_{pin}(t) = \pm V_p(f_o)\ e^{-\pi f_o t/Q} \sin(2\pi f_o t)$

where:

f_o = resonant frequency in Hz

t = time in seconds

Q = decay factor = 16 ±4

and $I_p\ (f_o)$ and $V_p\ (f_o)$ are given as a function of the resonant frequency as shown in FIGURE 5".

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FIGURE 5, title: Delete "IP (fo) and V_p (fo)" and substitute "$I_p\ (f_o)$ and $V_p\ (f_o)$".

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3.6.1, line 1: Delete "J-3546/USQ-74" and substitute "J-3546A/USQ-74A".

3.6.3.1c: Delete "(green)" and substitute "(white)".

3.6.3.1, line 12: Delete "J-3546/USQ-74" and substitute "J-3546A/USQ-74A".

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3.9.3.1, line 1: Delete "green" and substitute "white".

Preparing activity:
NAVY - EC

(Project No. 7025-N013(EC))

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3.2.1.11.1 Preamble. The preamble shall be the first five frames of every transmission and shall be a 2-tone signal consisting of a 605 Hz and a 2915 Hz tone. The 605-Hz tone shall be used for doppler correction and the 2915-Hz tone shall be used for synchronization. The 2915-Hz tone shall be phase shifted 180 degrees each successive frame enabling the receiving unit to detect frame transitions.

3.2.1.11.2 Preamble frame level. During the preamble the level of the 605-Hz tone shall be 12 ±1 dB above the level of a phase reference frame data subcarrier tone and the 2915-Hz tone shall be 6 ±1 dB above the level of a phase reference frame data subcarrier tone.

3.2.1.11.3 Phase reference frame. The phase reference frame shall immediately follow the preamble, shall be composed of the normal 16-tone composite, and shall serve as the phase reference for the first frame of the information segment.

3.2.1.11.4 Phase reference frame level. The phase reference frame shall be composed of the 16-tone composite signal. The difference in amplitude between the maximum level data subcarrier tone and the minimum level data subcarrier tone shall not exceed 1.5 dB.

NOTE: For 0 dB composite audio signal level across 600 ohms, the signal should have the following voltage levels:

Preamble:	605 Hz Doppler	= 0.692 vrms
	2915 Hz sync	= 0.346 vrms
Phase reference and information segment:	605 Hz doppler	= 0.356 volts rms
	data subcarrier tones	= 0.178 volts rms

3.2.1.12 Intermodulation distortion. Each intermodulation distortion product falling in the 500-Hz to 3050-Hz frequency band shall be a minimum of 38 dB below the level of either of any two equal amplitude data tones. In addition, for each 110-Hz frequency band with center frequencies of 935, 1045, 1155, 1265, 1375, 1485, 1595, 1705, 1815, 1925, 2035, 2145, 2255, 2365, and 2915 Hz, the peak value of the sum of all intermodulation products shall be a minimum of 40 dB below the level of two unmodulated tones at frequencies of 605 Hz and 2915 Hz. The combined power level of the two tones shall be 0 dBm ±0.5 dB, with the level of 2915 Hz tone 6 dB below the level of the 605 Hz tone. This requirement applies to all analog portions of both the receive processing circuits and the transmit processing circuits.

3.2.1.13 Bit error rate. The bit error rate (BER) performance in each sideband, USB and LSB, of the receive side of the CIU, when installed and functioning as a part of an AN/USQ-74 system, shall be not more than the BER shown in FIGURES 3 and 4. The BER requirement applies to an input signal range of from -6 to +3 dBm into 600 ohms without diversity.

3.2.2 Physical characteristics.

3.2.2.1 CV-3507A/USQ-74A. The case shall be designed to permit standard rack-mounting in accordance with EIA RS-310-C-77, using shelf support or hard-mounting to a horizontal surface. Mounting options shall be available by means of interchangeable brackets mounted to the sides of the case. The dimensions and weight of the unit without brackets attached shall be as follows:

	Limit
Maximum (max) height	23.5 centimeters (cm) (9.25 inches (in.))
Max width	43.2 cm (17 in.)
Max depth	48.3 cm (19 in.)
Max weight	31.8 kilograms (kg) (70 pounds (lbs))

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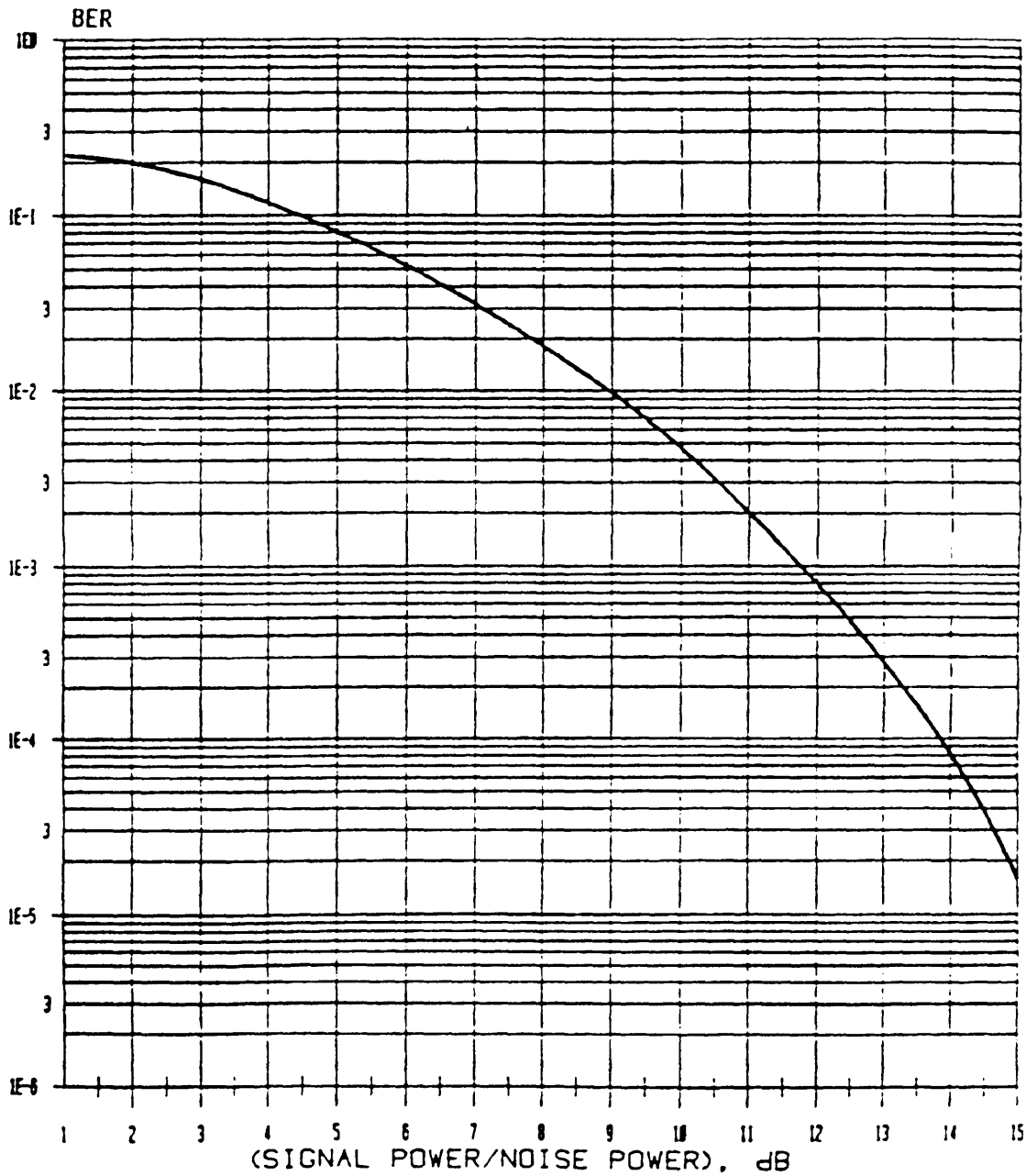


FIGURE 3. Bit error rate performance.
(fast rate)

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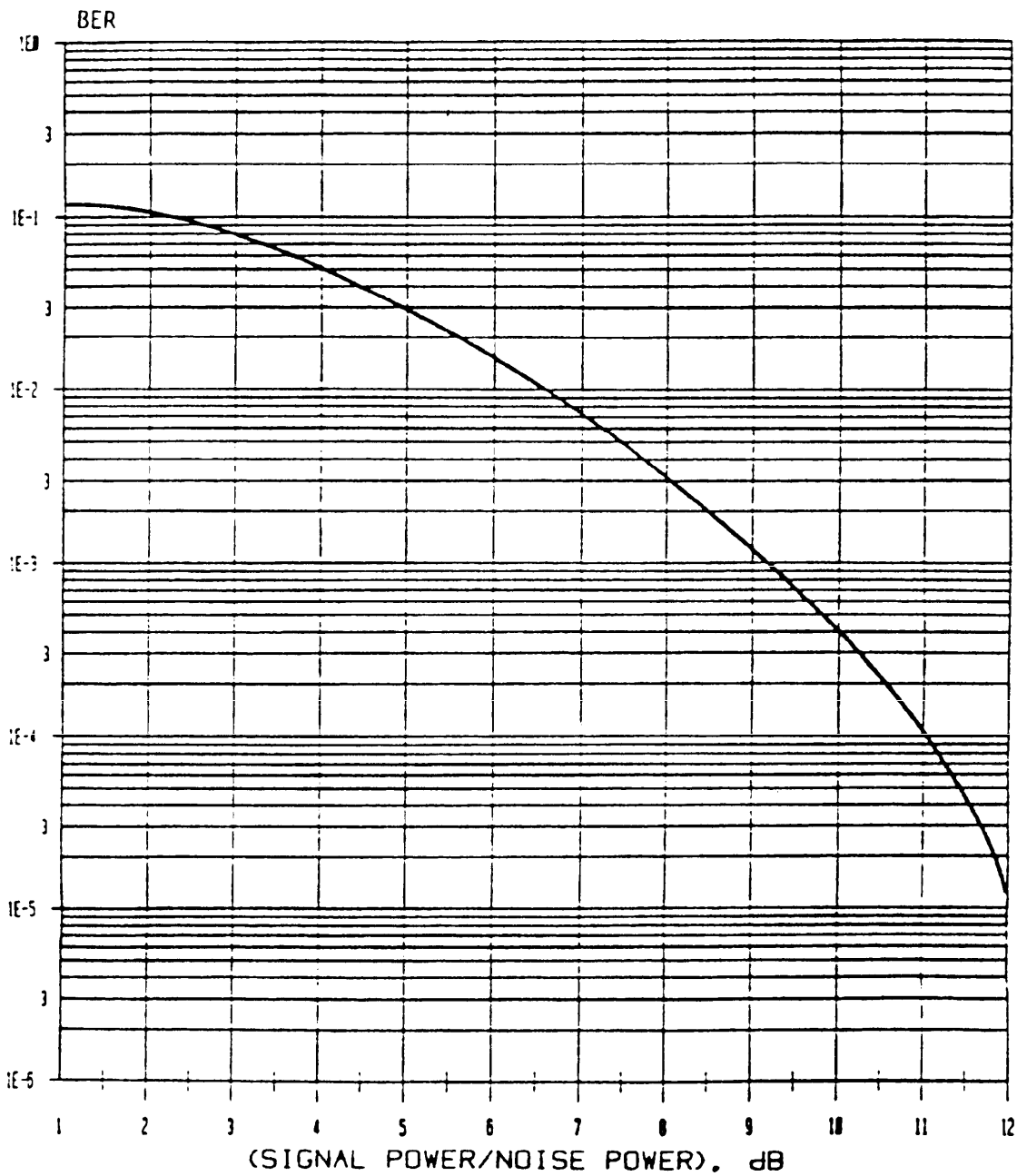


FIGURE 4. Bit error rate performance.
(slow rate)

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When the side-mounted brackets are attached to the case and the top and bottom brackets are attached, the overall dimensions increase to the following:

Max height	26.7 cm (10.5 in.)
Max width	48.3 cm (19 in.)

The unit shall be delivered without brackets attached. Shelf-mounting instructions shall be included in the operation and maintenance manual.

3.2.2.2 J-3546A/USQ-74A. The case shall be designed to permit 8-32 bolt mounting by drilling mounting holes which coincide with the chassis mounting holes. The dimensions and weight of the audio monitor unit shall be as follows:

Width (max)	Height (max)	Depth (max)	Weight (max)
5.25 in.	3.75 in.	6.5 in.	3 lb
13.3 cm	9.5 cm	17.5 cm	1.36 kg

3.2.3 Allen wrenches. If Allen wrenches are required to remove any component of the CIU during any maintenance action, the proper Allen wrenches shall be provided with each CIU. The Allen wrench(es) shall have suitable stowage mounting provisions inside the case.

3.2.4 Cases. The cases shall be constructed of aluminum.

3.2.4.1 Case bonding and grounding. Case bonding and grounding shall be accomplished by preparing mounting surfaces to provide metal contact between the case and the mounting bracket, and between the mounting bracket and the mounting surface. The area around each threaded insert of each case used to secure the mounting bracket to the case shall be free of paint and chemically treated to preserve the metal and provide a ground. Mounting brackets shall be similarly treated to insure bonding and grounding.

3.2.4.1.1 Grounding stud. A .375-in. (.9525 cm) diameter grounding stud shall be installed on the rear panel of the CIU. The grounding stud shall be supplied with nut and washer.

3.2.5 Construction.

3.2.5.1 Modular construction. Modular construction shall be in accordance with MIL-E-16400 except that the standard hardware program module requirements of MIL-STD-1378 shall not apply.

3.2.5.2 Circuit boards and wire wrapping. Printed circuit boards shall be used to the maximum extent possible. Wire wrapping shall not be used.

3.2.5.3 Hardware configuration. The CIU shall be wired to accommodate all interface features and requirements as specified in 3.1. The CIU interface configurations shall be changed by the plug-in addition, removal, or replacement of appropriate hardware modules without modification to any CIU hardware. The contractor shall provide all interface modules necessary for operation with the equipment specified in 3.1.

3.2.6 Human engineering. The CIU shall conform to the human engineering criteria of MIL-STD-1472.

3.2.7 Reliability requirement.

3.2.7.1 Quantitative reliability requirement. The CIU shall have an upper test limit mean-time-between-failure (MTBF) (θ_0 as defined by MIL-STD-781) of 6000 hours. The lower test limit MTBF (θ_1 as defined by MIL-STD-781) shall be 2000 hours.

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